

• **EXPRESSION DANS LES DIFFERENTS SYSTEMES DE COORDONNEES**

○ En coordonnées cartésiennes

$$\overrightarrow{\text{grad}} U = \frac{\partial U}{\partial x} \vec{e}_x + \frac{\partial U}{\partial y} \vec{e}_y + \frac{\partial U}{\partial z} \vec{e}_z$$

$$\text{div}(\vec{A}) = \frac{\partial a_x}{\partial x} + \frac{\partial a_y}{\partial y} + \frac{\partial a_z}{\partial z}$$

$$\overrightarrow{\text{rot}}(\vec{A}) = \left[\frac{\partial a_z}{\partial y} - \frac{\partial a_y}{\partial z} \right] \vec{e}_x + \left[\frac{\partial a_x}{\partial z} - \frac{\partial a_z}{\partial x} \right] \vec{e}_y + \left[\frac{\partial a_y}{\partial x} - \frac{\partial a_x}{\partial y} \right] \vec{e}_z$$

$$\Delta U = \frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2}$$

$$\Delta \vec{A} = \left[\frac{\partial^2 a_x}{\partial x^2} + \frac{\partial^2 a_x}{\partial y^2} + \frac{\partial^2 a_x}{\partial z^2} \right] \vec{e}_x + \left[\frac{\partial^2 a_y}{\partial x^2} + \frac{\partial^2 a_y}{\partial y^2} + \frac{\partial^2 a_y}{\partial z^2} \right] \vec{e}_y + \left[\frac{\partial^2 a_z}{\partial x^2} + \frac{\partial^2 a_z}{\partial y^2} + \frac{\partial^2 a_z}{\partial z^2} \right] \vec{e}_z$$

○ En coordonnées cylindriques

$$\overrightarrow{\text{grad}} U = \frac{\partial U}{\partial r} \vec{e}_r + \frac{1}{r} \frac{\partial U}{\partial \theta} \vec{e}_\theta + \frac{\partial U}{\partial z} \vec{e}_z$$

$$\text{div}(\vec{A}) = \frac{1}{r} \frac{\partial (r a_r)}{\partial r} + \frac{1}{r} \frac{\partial (a_\theta)}{\partial \theta} + \frac{\partial a_z}{\partial z}$$

$$\overrightarrow{\text{rot}}(\vec{A}) = \left[\frac{1}{r} \frac{\partial a_z}{\partial \theta} - \frac{\partial a_\theta}{\partial z} \right] \vec{e}_r + \left[\frac{\partial a_r}{\partial z} - \frac{\partial a_z}{\partial r} \right] \vec{e}_\theta + \left[\frac{1}{r} \frac{\partial (r a_\theta)}{\partial r} - \frac{1}{r} \frac{\partial a_r}{\partial \theta} \right] \vec{e}_z$$

$$\Delta U = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial U}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 U}{\partial \theta^2} + \frac{\partial^2 U}{\partial z^2}$$

$$\begin{aligned} \Delta(\vec{A}) = & \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial a_r}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 a_r}{\partial \theta^2} + \frac{\partial^2 a_r}{\partial z^2} - \frac{2}{r^2} \frac{\partial a_\theta}{\partial \theta} - \frac{a_r}{r^2} \right] \vec{e}_r \\ & + \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial a_\theta}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 a_\theta}{\partial \theta^2} + \frac{\partial^2 a_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial a_r}{\partial \theta} - \frac{a_\theta}{r^2} \right] \vec{e}_\theta \\ & + \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial a_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 a_z}{\partial \theta^2} + \frac{\partial^2 a_z}{\partial z^2} \right] \vec{e}_z \end{aligned}$$

○ En coordonnées sphériques

$$\overrightarrow{\text{grad}} U = \frac{\partial U}{\partial r} \vec{e}_r + \frac{1}{r} \frac{\partial U}{\partial \theta} \vec{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial U}{\partial \varphi} \vec{e}_\varphi$$

$$\text{div}(\vec{A}) = \frac{1}{r^2} \frac{\partial (r^2 a_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial (a_\theta \sin \theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial a_\varphi}{\partial \varphi}$$

$$\overrightarrow{\text{rot}}(\vec{A}) = \frac{1}{r \sin \theta} \left[\frac{\partial (a_\varphi \sin \theta)}{\partial \theta} - \frac{\partial a_\theta}{\partial \varphi} \right] \vec{e}_r + \left[\frac{1}{r \sin \theta} \frac{\partial a_r}{\partial \varphi} - \frac{1}{r} \frac{\partial (r a_\varphi)}{\partial r} \right] \vec{e}_\theta + \frac{1}{r} \left[\frac{\partial (r a_\theta)}{\partial r} - \frac{\partial a_r}{\partial \theta} \right] \vec{e}_\varphi$$

$$\Delta U = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial U}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial U}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 U}{\partial \varphi^2} = \frac{1}{r} \frac{\partial^2 (rU)}{\partial r^2} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial U}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 U}{\partial \varphi^2}$$

• **FORMULAIRE RELATIF AUX OPERATEURS**

$$dU = \overrightarrow{\text{grad}} (U(M)) \cdot d\vec{OM}$$

$$\text{div}(\overrightarrow{\text{grad}} U) = \Delta U$$

$$\overrightarrow{\text{rot}}(\overrightarrow{\text{grad}} U) = 0$$

$$\text{div}(\overrightarrow{\text{rot}} \vec{A}) = 0$$

$$\overrightarrow{\text{rot}}(\overrightarrow{\text{rot}} \vec{A}) = \overrightarrow{\text{grad}} (\text{div}(\vec{A})) - \Delta \vec{A}$$

$$\overrightarrow{\text{grad}} (UV) = V \overrightarrow{\text{grad}} (U) + U \overrightarrow{\text{grad}} (V)$$

$$\text{div} (U \vec{A}) = \vec{A} \cdot \overrightarrow{\text{grad}} (U) + U \text{div} (\vec{A})$$

$$\overrightarrow{\text{rot}} (U \vec{A}) = \overrightarrow{\text{grad}} (U) \times \vec{A} + U (\overrightarrow{\text{rot}} \vec{A})$$

$$\text{div} (\vec{A} \times \vec{B}) = \vec{B} \cdot (\overrightarrow{\text{rot}} \vec{A}) - \vec{A} \cdot (\overrightarrow{\text{rot}} \vec{B})$$

$$\overrightarrow{\text{rot}} (\vec{A} \times \vec{B}) = (\text{div} \vec{B}) \vec{A} - (\text{div} \vec{A}) \vec{B} + (\vec{B} \cdot \overrightarrow{\text{grad}}) \vec{A} - (\vec{A} \cdot \overrightarrow{\text{grad}}) \vec{B}$$

$$\overrightarrow{\text{grad}} (\vec{A} \cdot \vec{B}) = \vec{A} \times (\overrightarrow{\text{rot}} \vec{B}) + \vec{B} \times (\overrightarrow{\text{rot}} \vec{A}) + (\vec{A} \cdot \overrightarrow{\text{grad}}) \vec{B} + (\vec{B} \cdot \overrightarrow{\text{grad}}) \vec{A}$$